

刁恩茂

勇敢、创新、好奇、诚实、坚韧
分布式机器学习、高效机器学习、信号处理、人工智能

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教育

- 杜克大学 | 电气工程博士 05/2018 – 09/2023
- 哈佛大学 | 工程科学硕士 08/2016 – 05/2018
- 佐治亚理工学院 | 计算机科学学士 (最高荣誉) 08/2013 – 05/2016
- 佐治亚理工学院 | 电气工程学士 (最高荣誉) 08/2012 – 05/2016

经历

- 创始人、首席执行官兼首席科学家**
灵机一梦 09/2024 – 现在
 - 创立一家 AI 公司, 构建面向个人 AI 助手的 AI 操作系统 (AIOS), 使 AI 模型能够在以消费级显卡为主的通用硬件上进行受存储受限的训练与推理
 - 研发与 PyTorch 兼容的运行系统, 实现 AI 系统在 GPU、CPU 内存及硬盘之间的存储管理
- 博士后**
杜克大学 09/2023 – 01/2024
 - 开发用于分布式机器学习和通用人工智能的高效协作方法
 - 受 Prof. Vahid Tarokh 和 Prof. Jie Ding 指导
- 研究助理**
杜克大学 05/2018 – 09/2023
 - 研究分布式机器学习, 开发联邦学习与辅助学习框架
 - 研究高效能机器学习, 提出基于深度学习的数据与模型压缩方法
 - 受 Prof. Vahid Tarokh 和 Prof. Jie Ding 指导
- 哈佛大学** 08/2016 – 05/2018
 - 从统计效率角度研究机器学习的理论极限
 - 开发高效算法计算非线性模型自由度, 将交叉验证计算量降低数个数量级
 - 受 Prof. Vahid Tarokh 和 Prof. Jie Ding 指导
- 佐治亚理工学院** 05/2014 – 05/2016
 - 开发用于移动应用中的单音音高转录的旋律提取算法
 - 在 FPGA 上以 Verilog 开发细胞神经网络
 - 受 Dr. Udit Gupta, Prof. Elliot Moore 和 Prof. Hyesoon Kim 指导
- 应用科学家**
亚马逊 01/2022 – 09/2022
 - 研究半监督与个性化联邦学习, 并实现其在 Alexa 设备中的应用
 - 开发基于 PyTorch 的内部研究开发流程
 - 与多位科学家协作研究联邦学习在公平性、异步性、辅助信息与模型压缩方面的方向
 - 受 Prof. Jie Ding 和 Dr. Tao Zhang 指导
- 项目管理**
杜克大学 Data+ 项目 05/2019 – 08/2019

– 领导由本科生组成的研究团队，开发语音情感识别与情感语音生成算法

• **教学助理**

杜克大学

08/2019 – 05/2020

– 指导学生学习并实现深度学习与音频信号处理算法

佐治亚理工学院

05/2015 – 08/2015

– 指导学生学习 FPGA、示波器操作，并使用 VHDL 构建处理器

论文

- X. Chen, T. Du, C. Xu, F. Zhuang, L. Zhong, G. Muntean, **E. Diao**, **FedOBP: Federated Optimal Brain Personalization through Cloud-Edge Element-wise Decoupling**, *Arxiv*
- T. Wang, J. Lu, X. Wang, H. Huang, S. Yao, Z. Hu, X. Chen, **E. Diao**, **RPS: Information Elicitation with Reinforcement Prompt Selection**, *Arxiv*
- Z. Sheng, J. Zhang, **E. Diao**, **Toward Unifying Group Fairness Evaluation from a Sparsity Perspective**, *Arxiv*
- Z. Li, H. Hu, S. Lim, X. Li, F. Gao, **E. Diao**, Z. Ding, M. Vazirgiannis, H. Bostrom, **A Kinetic-Energy Perspective of Flow Matching**, *ICML 2026 (Spotlight)*
- Y. Gu, X. Liang, J. Zhao, **E. Diao**, **OBCache: Optimal Brain KV Cache Pruning for Efficient Long-Context LLM Inference**, *ICML 2026*
- Z. Wang, **E. Diao**, Qi Le, P. Wang, M. Lee, S. Yeh, E., H., L. Yang, **From Local to Global: Revisiting Structured Pruning Paradigms for Large Language Models**, *ACL 2026*
- Z. Wang, **E. Diao**, Qi Le, P. Wang, G. Wang, M. Lee, S. Yeh, L. Yang, **Think Before You Prune: Self-Reflective Structured Pruning for Reasoning Language Models**, *DAC 2026*
- Z. Guo, **E. Diao**, C. Yang, C. Shi, **Graph Tokenization for Bridging Graphs and Transformers**, *ICLR 2026*
- X. Wang, J. Du, A. Khan, Qi Le, **E. Diao**, J. Zhou, J. Ding, A. Anwar, **Beyond Expectations: Quantile-Guided Alignment for Risk-Calibrated Language Models**, *NeurIPS 2025 (Spotlight)*
- S. Moushegian, S. Wu, **E. Diao**, J. Ding, T. Banerjee, V. Tarokh, **Robust Score-Based Quickest Change Detection**, *IEEE Transactions on Information Theory*
- Q. Le, **E. Diao**, Z. Wang, X. Wang, J. Ding, L. Yang, A. Anwar, **Probe Pruning: Accelerating LLMs through Dynamic Pruning via Model-Probing**, *ICLR 2025*
- X. Wang, Q. Le, A. Ahmed, **E. Diao**, Y. Zhou, N. Baracaldo, J. Ding, A. Anwar, **MAP: Multi-Human-Value Alignment Palette**, *ICLR 2025 (Oral)*
- X. Wang, **E. Diao**, Q. Le, J. Ding, A. Anwar, **AID: Adaptive Integration of Detectors for Safe AI with Language Models**, *NAACL 2025*
- Q. Le, **E. Diao**, X. Wang, V. Tarokh, J. Ding, A. Anwar, **DynamicFL: Federated Learning with Dynamic Communication Resource Allocation**, *IEEE BigData 2024 (Best Student Paper)*
- Y. Zhe, **E. Diao**, **ESC: Efficient Speech Coding with Cross-Scale Residual Vector Quantized Transformers**, *EMNLP 2024*
- Y. Zhang, **E. Diao**, D. Huston, T. Xia, **A PixelCNN Based Method for Rough Surface Clutter Reduction in GPR B-scan Images**, *IEEE Radar Conference*
- Y. Zhang, **E. Diao**, D. Huston, T. Xia, **A Data Efficient Deep Learning Method for Rough Surface Clutter Reduction in GPR Images**, *IEEE Transactions on Geoscience and Remote Sensing*

- **E. Diao**, T. Banerjee, V. Tarokh, **Large Deviation Analysis of Score-based Hypothesis Testing**, *IEEE Access*
- **E. Diao**, Q. Le, S. Wu, X. Wang, A. Anwar, J. Ding, V. Tarokh, **ColA: Collaborative Adaptation with Gradient Learning**, *arXiv*
- S. Wu, **E. Diao**, T. Banerjee, J. Ding, V. Tarokh, **Quickest Change Detection for Unnormalized Statistical Models**, *IEEE Transactions on Information Theory*
- K. Varma, **E. Diao**, T. Roosta, J. Ding, T. Zhang, **Once-for-All Federated Learning: Learning From and Deploying to Heterogeneous Clients**, *KDD 2023 Workshop*
- S. Wu, **E. Diao**, T. Banerjee, J. Ding, V. Tarokh, **Robust Quickest Change Detection for Unnormalized Models**, *UAI 2023*
- **E. Diao**, T. Eric, J. Ding, Z. Tao, **Semi-Supervised Federated Learning for Keyword Spotting**, *ICME 2023*
- **E. Diao**, G. Wang, J. Zhang, Y. Yang, J. Ding, V. Tarokh, **Pruning Deep Neural Networks from a Sparsity Perspective**, *ICLR 2023*
- S. Wu, **E. Diao**, T. Banerjee, J. Ding, V. Tarokh, **Score-based Change Point Detection for Unnormalized Models**, *AISTATS 2022*
- Q. Le, **E. Diao**, X. Wang, A. Anwar, V. Tarokh, J. Ding, **Personalized Federated Recommender Systems with Private and Partially Federated AutoEncoders**, *Asilomar 2022*
- L. Collins, **E. Diao**, T. Roosta, J. Ding, T. Zhang, **PerFedSI: A Framework for Personalized Federated Learning with Side Information**, *NeurIPS 2022 Workshop*
- **E. Diao**, J. Ding, V. Tarokh, **SemiFL: Communication Efficient Semi-Supervised Federated Learning with Unlabeled Clients**, *NeurIPS 2022*
- **E. Diao**, J. Ding, V. Tarokh, **GAL: Gradient Assisted Learning for Decentralized Multi-Organization Collaborations**, *NeurIPS 2022*
- S. Wu, **E. Diao**, K. Elkhailil, J. Ding, V. Tarokh, **Score-based Hypothesis Testing for Unnormalized Models**, *IEEE Access*
- **E. Diao**, V. Tarokh, J. Ding, **Decentralized Multi-Target Cross-Domain Recommendation for Multi-Organization Collaborations**, *arXiv*
- **E. Diao**, J. Ding, V. Tarokh, **Multimodal Controller for Generative Models**, *CVMI 2022*
- M. Mohammadreza, **E. Diao**, V. Tarokh, B. Andrew, **Emulating Spatio-Temporal Realizations of Three-Dimensional Isotropic Turbulence via Deep Sequence Learning Models**, *AAAI 2022 Workshop*
- M. Mohammadreza, **E. Diao**, V. Tarokh, B. Andrew, **Dimension Reduced Turbulent Flow Data From Deep Vector Quantizers**, *Journal of Turbulence*
- M. Mohammadreza, **E. Diao**, V. Tarokh, B. Andrew, **A Physics-Informed Vector Quantized Autoencoder for Data Compression of Turbulent Flow**, *DCC 2021*
- **E. Diao**, J. Ding, V. Tarokh, **HeteroFL: Computation and Communication Efficient Federated Learning for Heterogeneous Clients**, *ICLR 2021*
- J. Ding, **E. Diao**, V. Tarokh, **On Statistical Efficiency in Learning**, *IEEE Transactions on Information Theory*
- J. Wang, M. Xue, R. Culhane, **E. Diao**, J. Ding, V. Tarokh, **Speech Emotion Recognition with Dual-Sequence LSTM Architecture**, *ICASSP 2020*

- **E. Diao, J. Ding, V. Tarokh, DRASIC: Distributed Recurrent Autoencoder for Scalable Image Compression, *DCC 2020***
- **S. Wu, E. Diao, J. Ding, V. Tarokh, Deep Clustering of Compressed Variational Embeddings, *DCC 2020***
- **E. Diao, J. Ding, V. Tarokh, Restricted Recurrent Neural Networks, *IEEE BigData 2019***
- **J. Ding, E. Diao, V. Tarokh, A Penalized Method for the Predictive Limit of Learning, *ICASSP 2018***

奖项

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|---|---------|
| • Student Travel Award IEEE BigData 2019 | 12/2019 |
| • ECE Senior Scholar Award Georgia Institute of Technology | 04/2016 |
| • President Undergraduate Research Award Georgia Institute of Technology | 07/2015 |